USD/CRC · MONEX — the calendar strategy: dynamics, rule logic & VWAP-priced results

2,663 sessions · 2014-12-08 → 2026-06-19 · long/short USD · $1,000,000 per trade · **priced VWAP-to-VWAP**, net of 0.65 CRC round-trip · no technical indicators

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The recommended strategy is a **calendar (quincena) rule**: short USD only into Costa Rica's mid-month IVA / payroll deadline (within 6 business days of it, or equivalently day-of-month 5–15), long the rest of the month. Priced at the session VWAP net of cost it earns **$125k/yr at Sharpe 3.0** on just 24.2 trades a year, with an **out-of-sample Sharpe of 2.91** ($152k/yr on the 40% held out after 2021-11-18) and a worst year of $34k — positive in all 12 years. A **dynamic trailing-stop exit** (A5) enhances it further — to $128k/yr at Sharpe 3.22, or Sharpe 3.91 with a hard floor that nearly halves the drawdown — **lifting P&L both in- and out-of-sample**. Every number below is VWAP-priced; secondary signals are kept in tabs but default to this rule.

$125k
calendar rule / yr
$1M/trade · VWAP-priced

3.0
Sharpe (full sample)
only 24.2 trades/yr

2.91
out-of-sample Sharpe
OOS $152k/yr · 40% held out

3.91
Sharpe with exit overlay
trailing stop cuts DD to $-26k (A5)

$34k
worst calendar year
positive in all 12 years

29 vs 18 M
volume: colón up vs down
USD sellers trade in size

PART A · THE RECOMMENDED STRATEGY — CALENDAR / QUINCENA RULE

A1. The rule & the numbers ($1M per trade, VWAP-priced, net of cost)

VERSION	PER YEAR	SHARPE	TRADES/YR	WIN %	MAX DD	NOTE
Base quincena (short if day ≤ 15)	$87k	2.05	24.2	56.2	$-62k	crude ancestor
Refined (short days 5–15, long otherwise)	$124k	2.96	24.2	59.5	$-50k	fixed window
Refined + slow-volume sizing	$103k	3.37	19.9	59.2	$-37k	best risk-adjusted
Real calendar (≤6 bd to IVA/quincena deadline) RECOMMENDED	$125k	3.0	24.2	59.8	$-48k	deadline-anchored

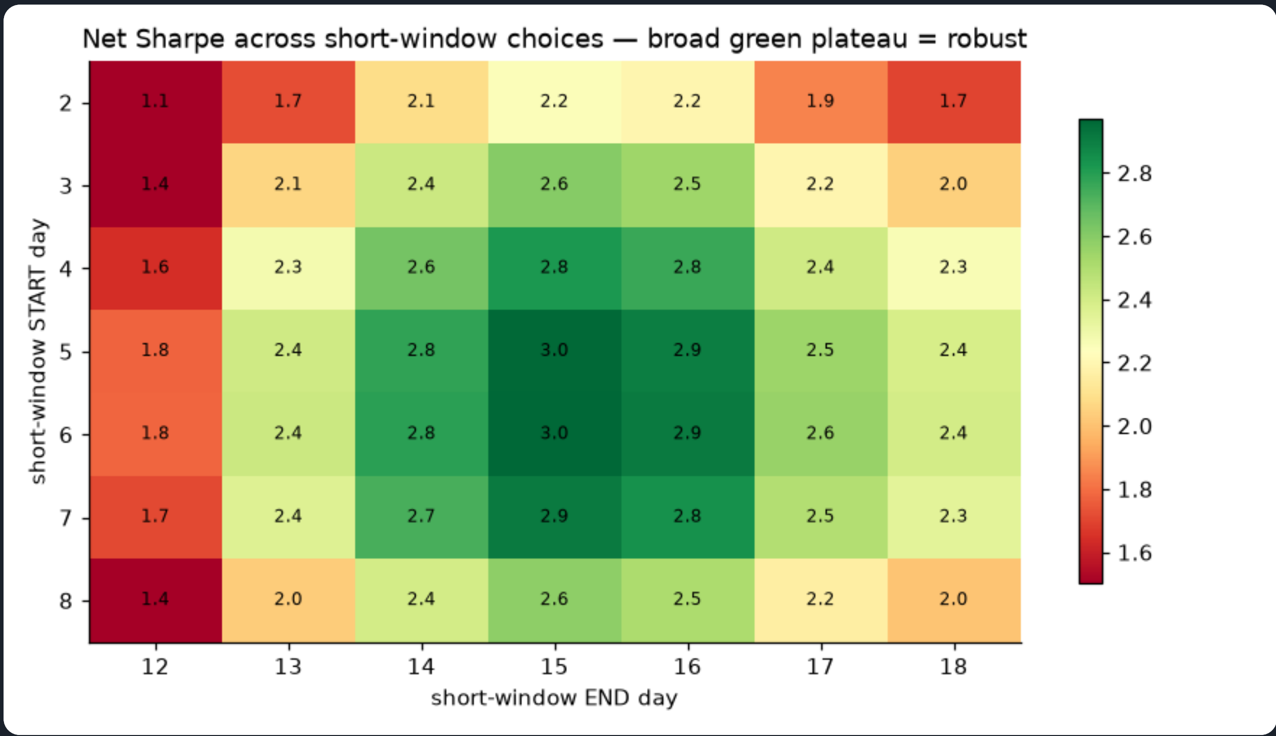
Exact logic (the trading calendar)

```
≤ 6 business days before the IVA/quincena deadline (through it) → SHORT USD
   (firms sell USD to raise colones for the 15th tax + payroll → colón
strengthens)
otherwise → LONG USD (supply fades, USD drifts up)
optional: trim size to 50% when a slow 20-day volume regime disagrees
```

Two equivalent readings of the same edge. The **fixed** version keys off the raw day number (days 1–4 LONG, 5–15 SHORT, 16–end LONG); the **real-calendar** version anchors the short window to Costa Rica's actual IVA (D-104) / mid-month quincena deadline — the 15th, rolled to the next business day — so it shifts with weekends and holidays. Anchoring to the true deadline earns $125k/yr at Sharpe 3.0 (vs $124k / 2.96 for the fixed window) at the same 24.2 trades/yr, and — see B2b — the next-day move is sharpest measured in *days-to-deadline*, confirming the cash-flow mechanism.

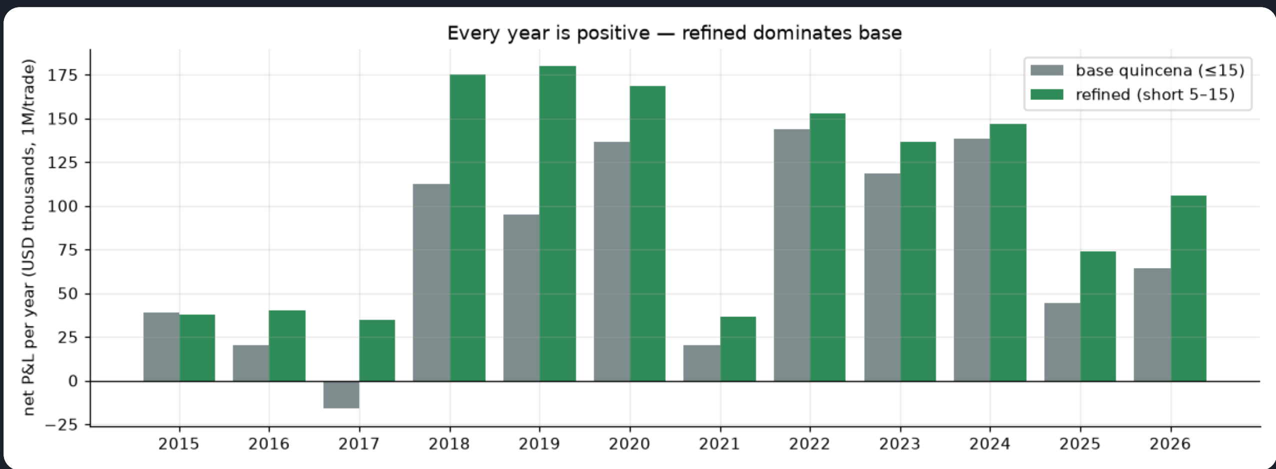
A2. Is it overfit? Window sensitivity & every-year P&L

Net Sharpe across every choice of the short-window start & end



The result is a broad green plateau (Sharpe 1.1–2.97), not a lucky point — days 5→15 is simply the natural mid-month supply window, and neighbours all work.

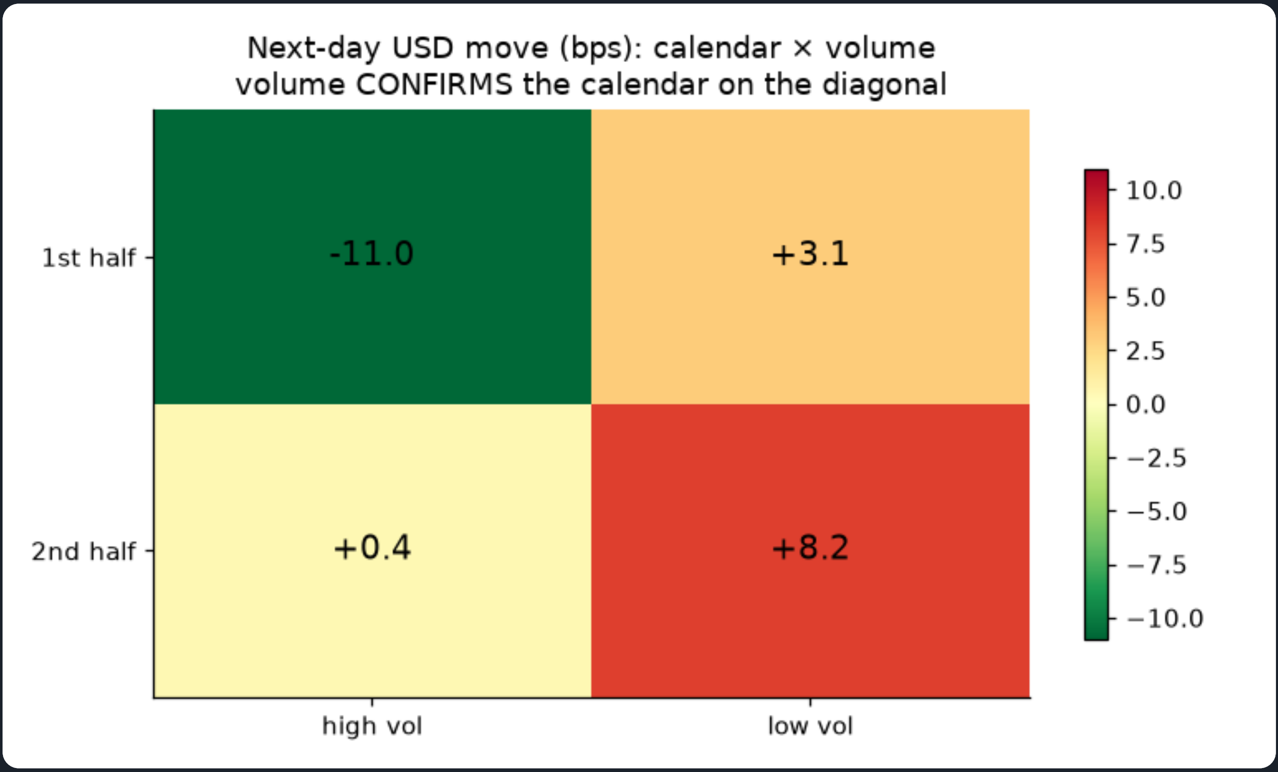
Net P&L by year — base vs refined ($1M/trade, VWAP-priced)



Positive in all 12 calendar years (worst $34k), including the 2015–17 pegged years where the volume signal was dead. The refined version dominates the base almost every year.

A3. How volume fits in — confirmation, not a daily signal

Next-day USD move: calendar × volume (bps)



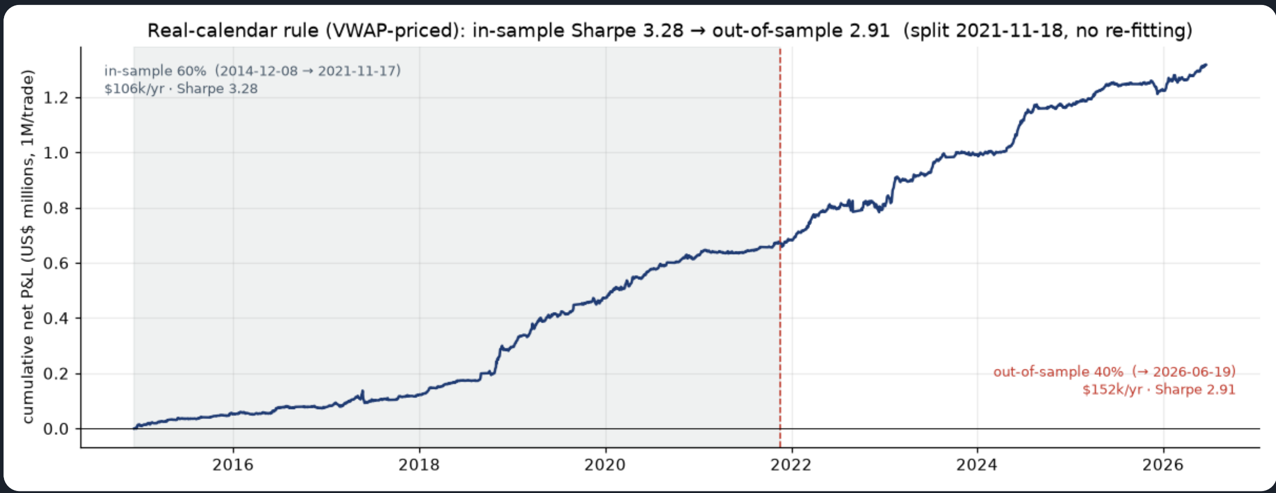
Volume CONFIRMS the calendar on the diagonal: first-half + high volume is strongly USD-down (sell USD), second-half + low volume strongly USD-up (buy USD). But trading volume day-by-day adds turnover that costs more than it adds — so we fold it in as a SLOW regime filter that only trims size, which is what lifts the Sharpe to 3.37 and cuts the drawdown to $-37k.

A4. Sample vs out-of-sample — the edge is not just the early years

> The rule is a fixed heuristic with **no fitted parameters**, but we still hold out the last 40% of history (everything after **2021-11-18**) to show the edge is not a property of the calmer, more-pegged early years. Trained-window intuition only; nothing is re-fit on the test window.

RULE (CHRONOLOGICAL 60/40 SPLIT, NO RE-FITTING)	IS / YR	IS SHARPE	OOS / YR	OOS SHARPE
Real calendar (≤6 bd)	$106k	3.28	$152k	2.91
Refined (short 5–15)	$108k	3.33	$147k	2.79

Real-calendar rule (VWAP-priced): in-sample vs out-of-sample equity



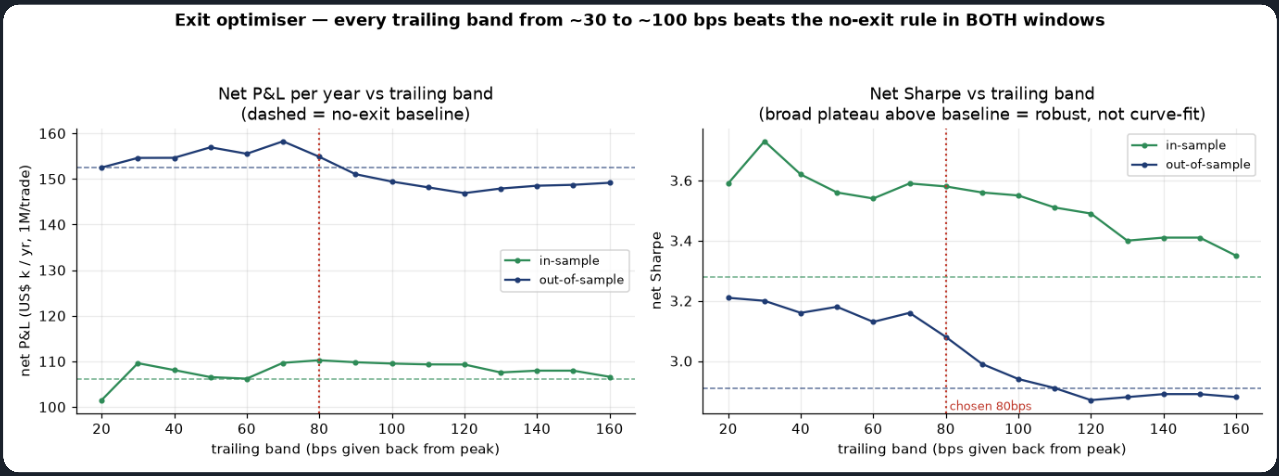
In-sample 2014-12-08 → 2021-11-17 (Sharpe 3.28) and out-of-sample → 2026-06-19 (Sharpe 2.91, $152k/yr). The out-of-sample slope is if anything steeper — the post-2021 float is where the cash-flow cycle is most tradeable.

A5. Enhancing the rule with a dynamic exit (stop-loss / take-profit)

The calendar rule is **always in the market** (short into the deadline, long otherwise). Because the edge **front-loads** — the colón strengthens *into* the deadline and reverts after — a trade often peaks mid-hold and then gives the move back. So we overlay a **trailing stop**: within each trade, once the running P&L gives back 80 bps from its best level, we bank it and stay flat until the calendar opens the next trade. This **never changes the entry logic** — it can only remove the losing tail of a trade. Both parameters are optimised on the **in-sample 60% only**; the out-of-sample 40% below is untouched.

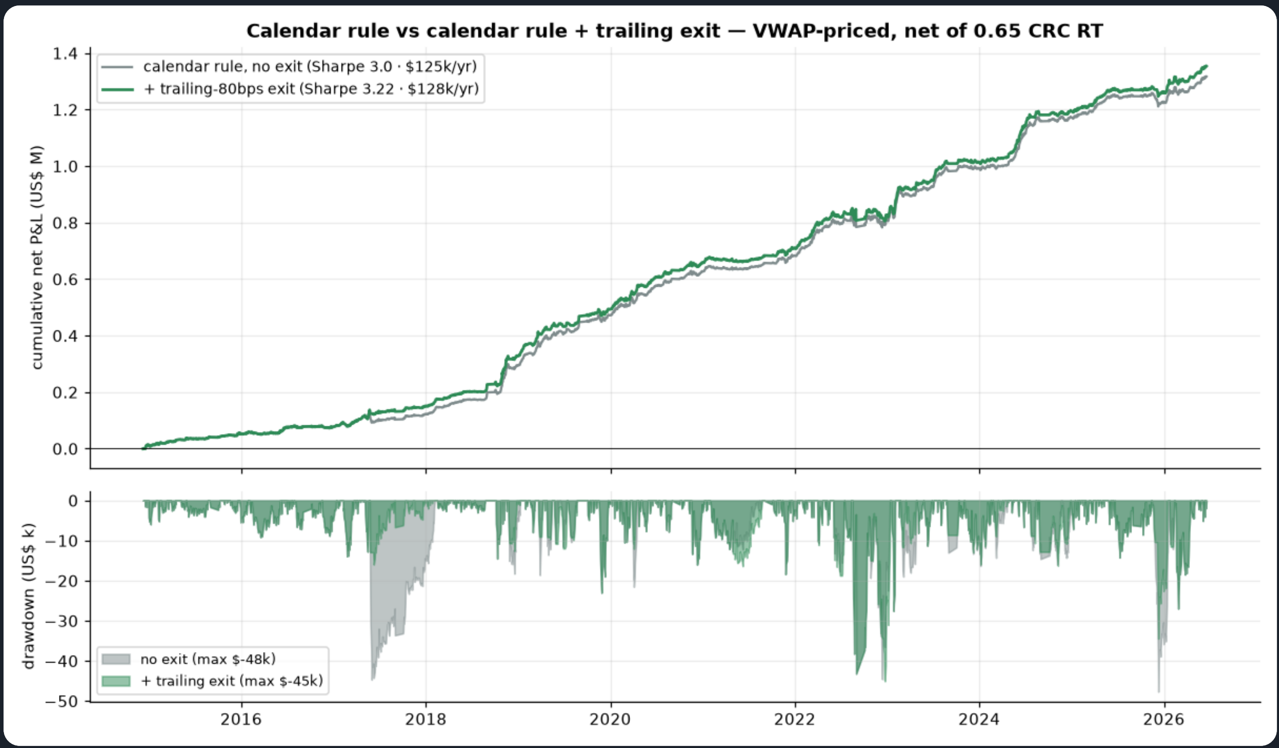
CALENDAR RULE · $1M/TRADE · VWAP-PRICED, NET COST	FULL / YR	SHARPE	MAX DD	OOS / YR	OOS SHARPE
No exit — always in the market the rule from A1–A4	$125k	3.0	$-48k	$152k	2.91
+ trailing-stop exit (80 bps) RECOMMENDED tuned on in-sample P&L; biggest raw P&L	$128k	3.22	$-45k	$155k	3.08
+ trailing (30) & hard floor (60) bps tuned on in-sample Sharpe; best risk-adjusted	$126k	3.91	$-26k	$153k	4.09

Exit optimiser — net P&L and Sharpe across the trailing band



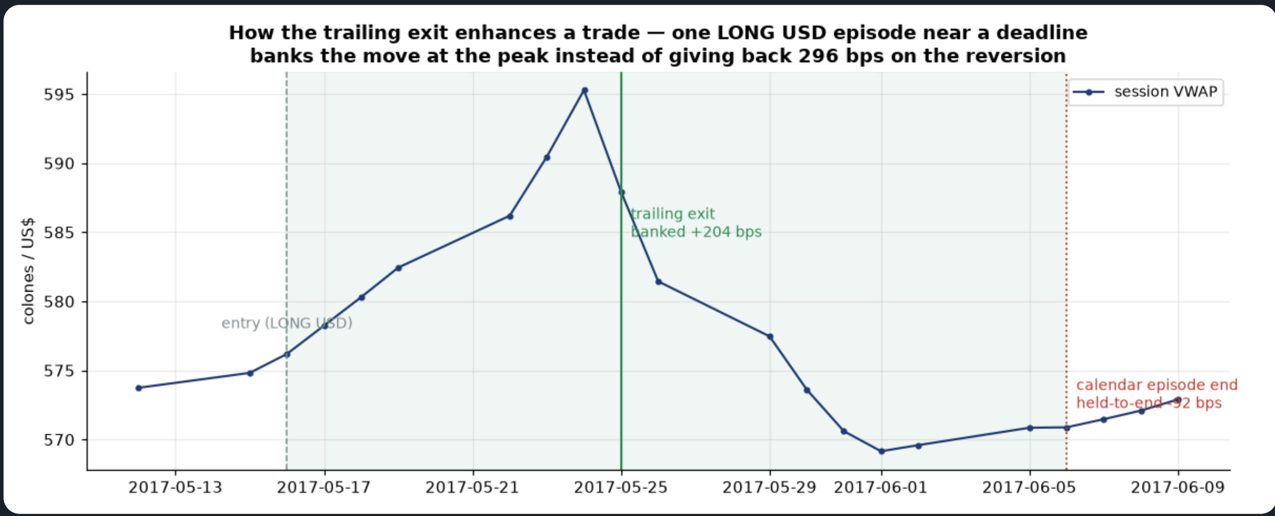
This is the search itself. Every trailing band from ~30 to ~100 bps beats the no-exit rule (dashed) in **both** the in-sample and out-of-sample windows on Sharpe, and lifts in-sample P&L across the whole range — a broad plateau, not a single lucky point (the same robustness test we applied to the calendar window in A2). The chosen 80 bps maximises in-sample P&L/yr.

Calendar rule vs calendar rule + trailing exit — equity & drawdown



Adding the exit lifts full-sample P&L from $125k/yr to $128k/yr (Sharpe 3.0 → 3.22) and — with the hard floor variant — cuts the worst drawdown from $-48k to $-26k. The green curve sits above the grey throughout, most visibly through the deep 2017–18 drawdown.

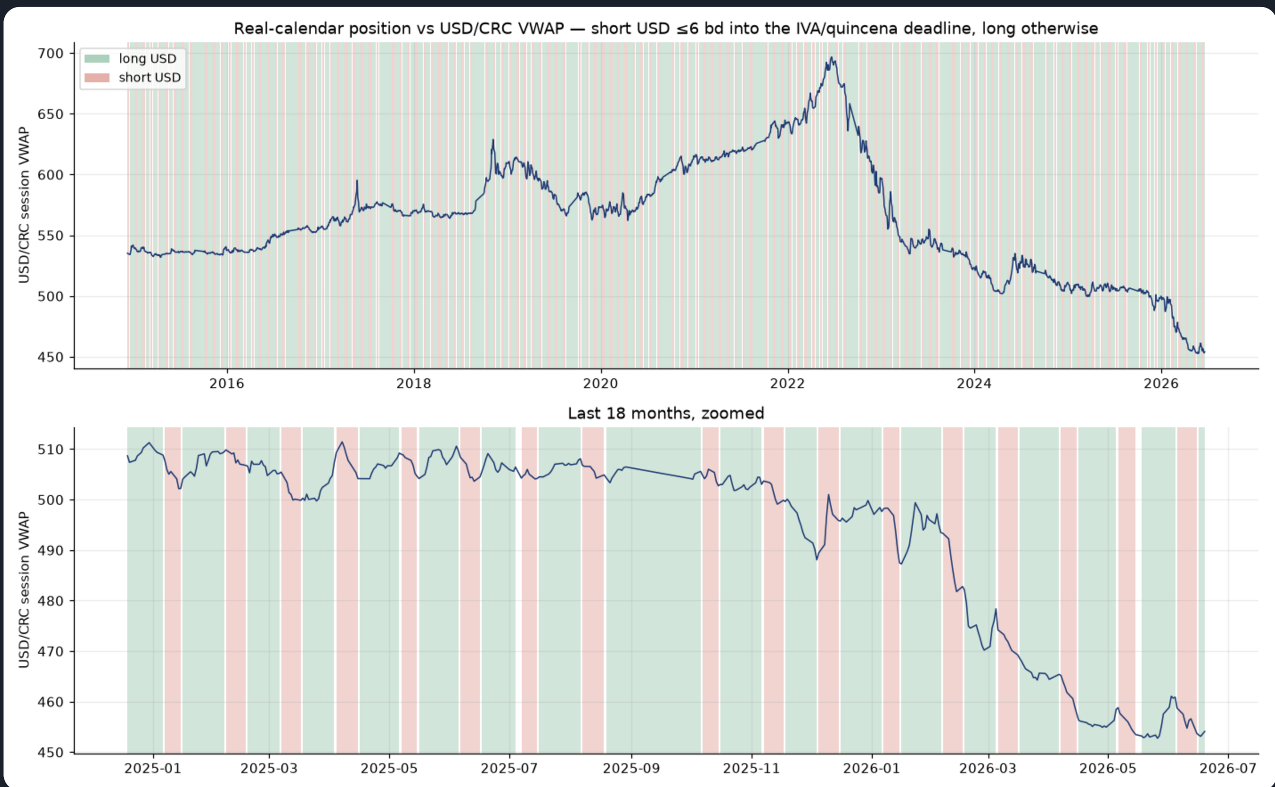
How the exit enhances one trade — bank the move at the peak, skip the reversion



A concrete long-USD trade: the position banks +204 bps at the trailing exit near the peak, instead of holding to the calendar's episode end and watching the same trade turn into a -92 bps result on the post-deadline reversion — that one exit is worth ~296 bps. The overlay does this systematically across trades; it is the mechanism, on a single trade.

A6. The position against the price

USD/CRC session VWAP shaded by the recommended calendar position



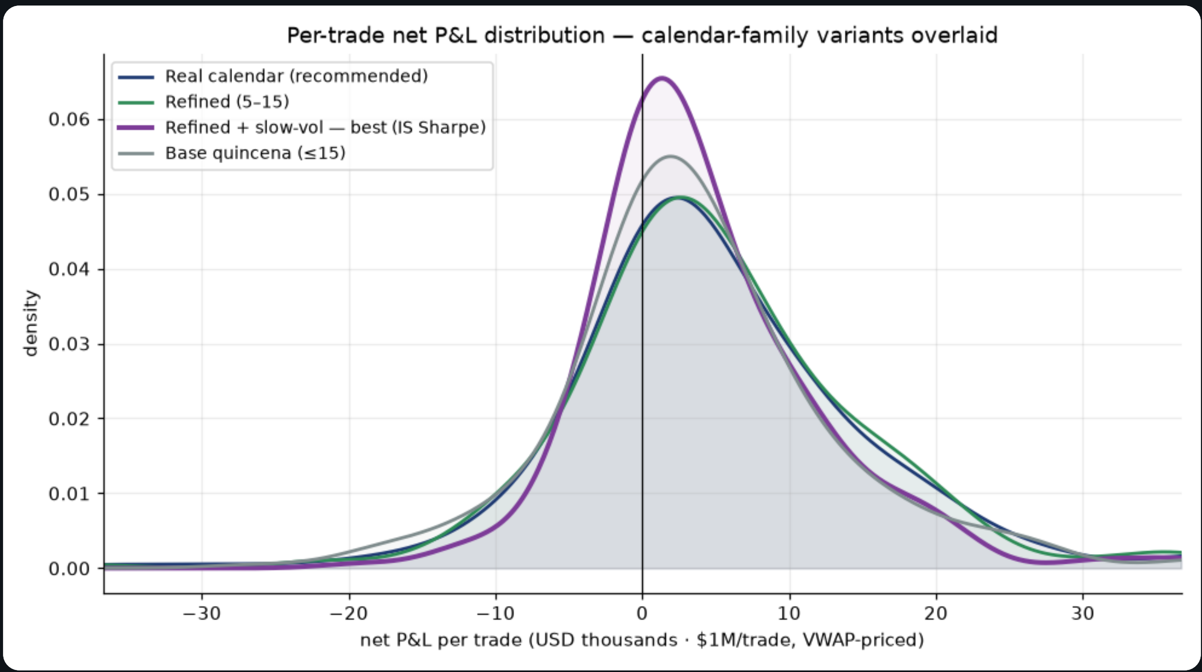
Green = long USD, red = short USD (short ≤6 business days into the IVA/quincena deadline). The rule is always in a position — there is no flat/no-trade state — so every session carries one shade or the other. The short (red) bands cluster mid-month, exactly where the cash-flow supply surge lands.

A7. Return distributions — how the variants compare

Net P&L **per trade** (per directional roundtrip), in USD at $1M/trade, **VWAP-to-VWAP** — the P&L a desk actually books at each exit, not a day-by-day mark. Tabs isolate one variant; the overlay puts all four on a shared x-range so the shapes compare directly. Variants are ranked by **in-sample Sharpe** (first 60% of history), *not* headline P&L: on that measure **Refined + slow-vol** wins and carries the best badge — it trims the fat losing trades, so its per-trade spread is tightest even where the calendar rule books more total dollars.

Overlay (all four)

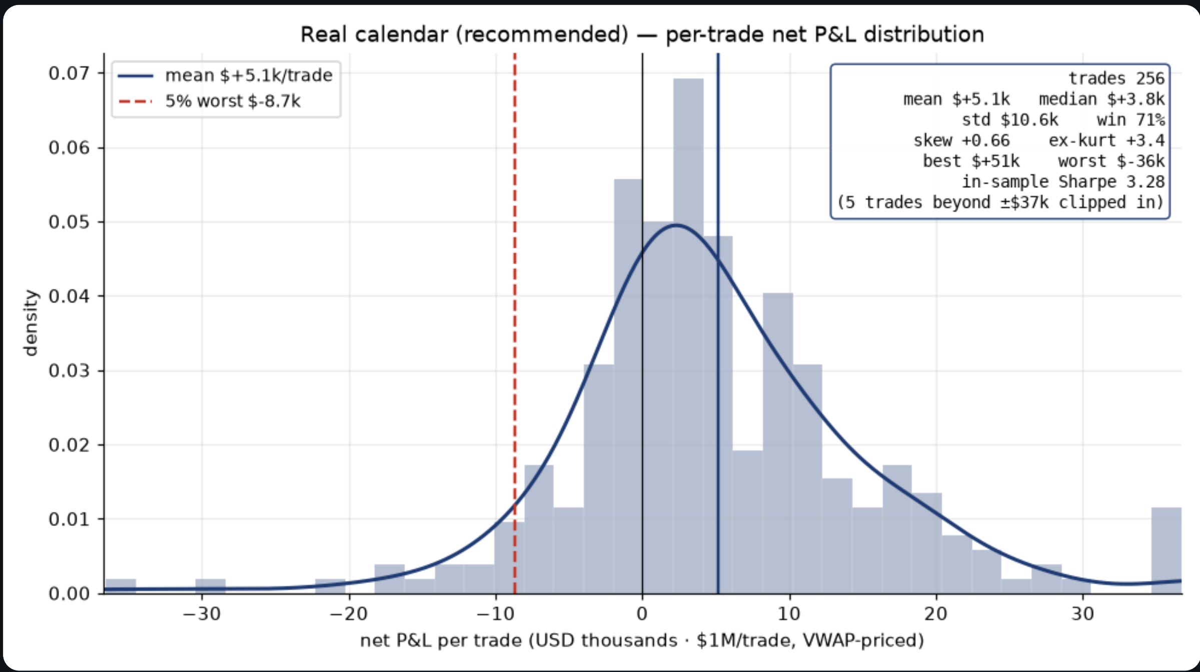
All four calendar-family variants, per-trade P&L overlaid



The slow-vol variant is the most peaked (tightest per-trade spread); the calendar rule and the fixed refined window sit almost on top of each other, both shifted right of the crude base rule.

Real calendar

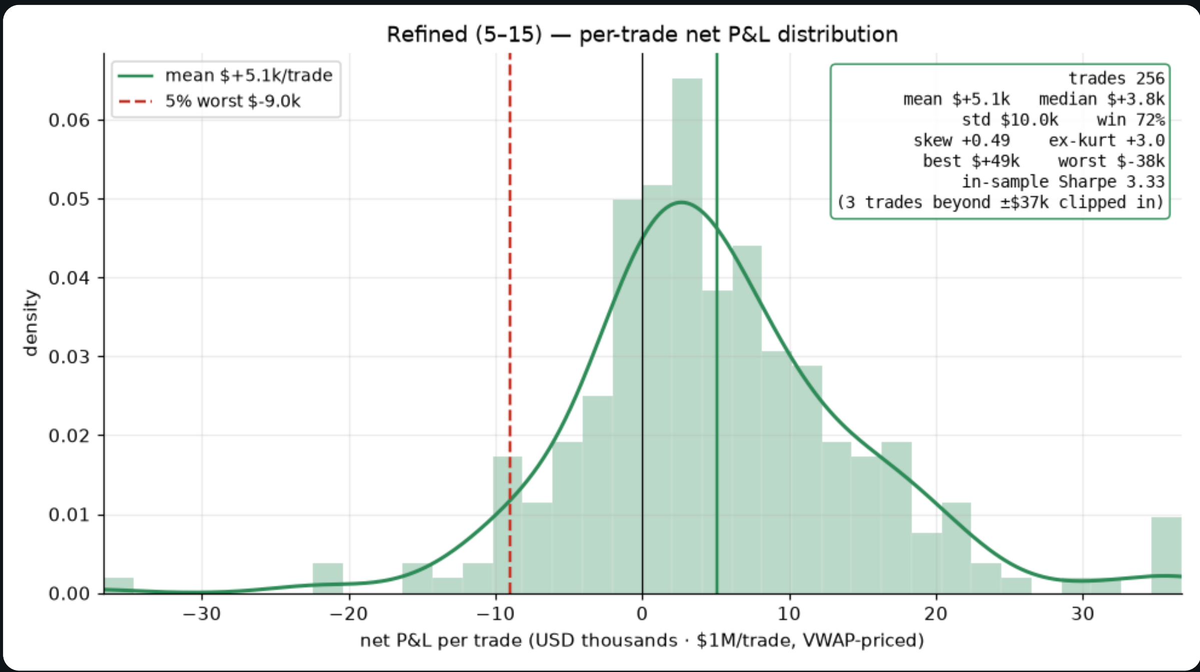
Real calendar (recommended) — per-trade net P&L distribution



The recommended rule. 256 trades · mean $+5.1k/trade, median $+3.8k, win 71% · std $10.6k · skew +0.66 · in-sample Sharpe 3.28.

Refined (5–15)

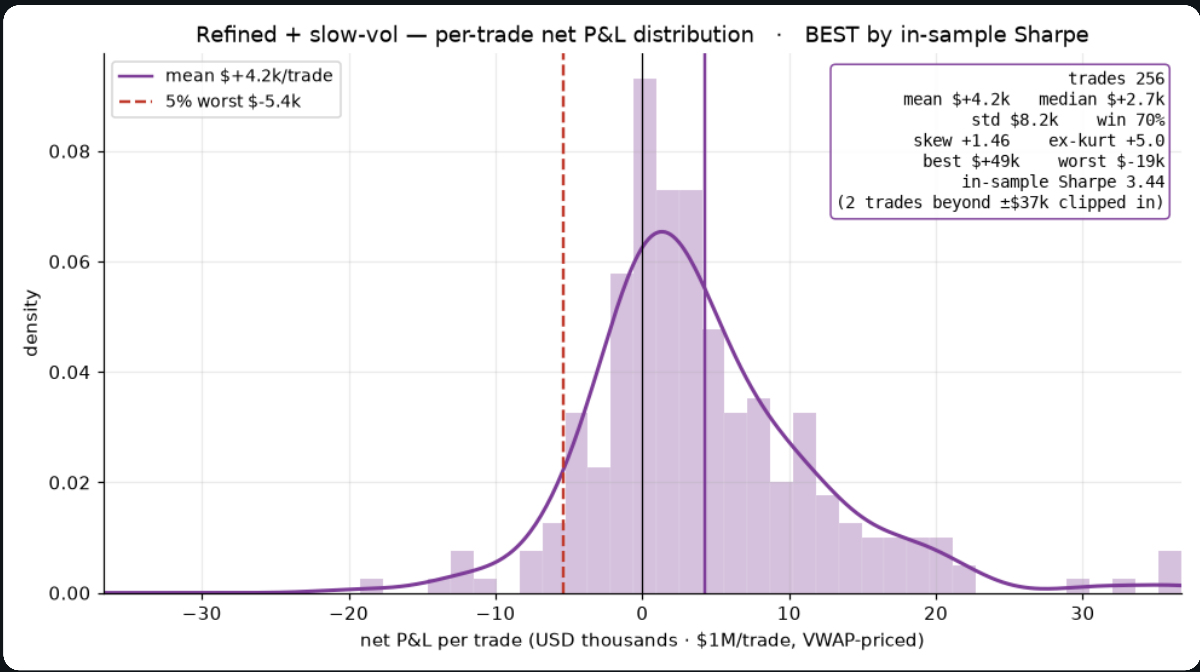
Refined quincena (5–15) — per-trade net P&L distribution



The fixed day-of-month twin of the calendar rule — nearly the same trade population. 256 trades · mean $+5.1k/trade, median $+3.8k, win 72% · std $10.0k · skew +0.49 · in-sample Sharpe 3.33.

Refined + slow-vol

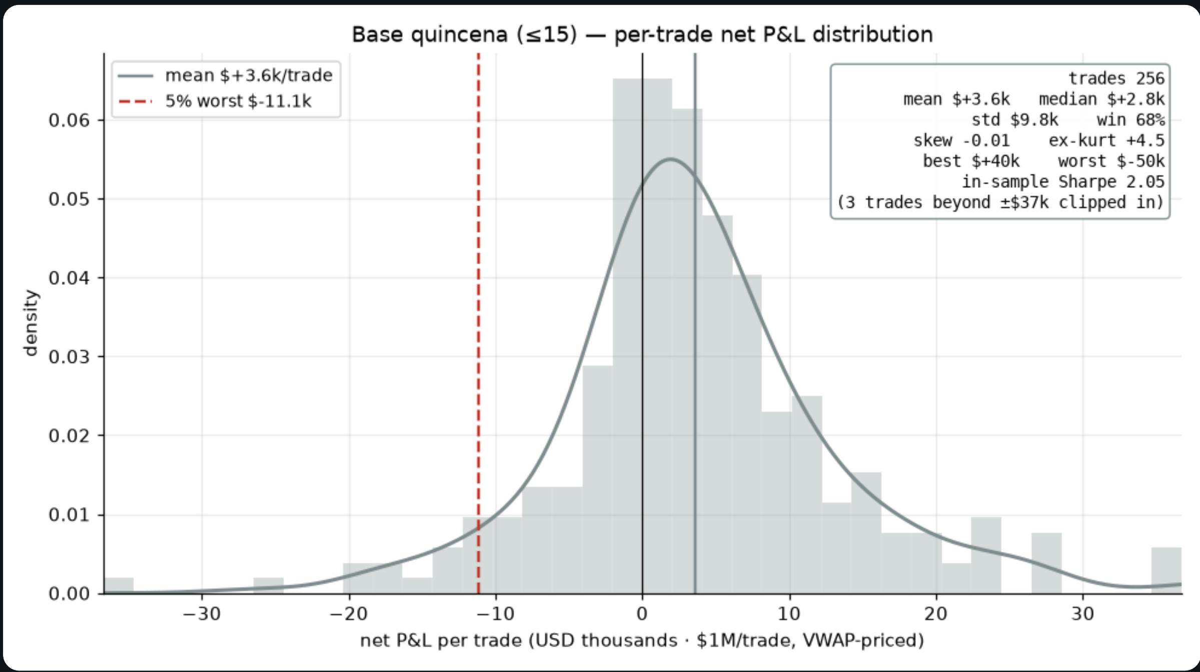
Refined + slow-vol sizing — per-trade net P&L distribution



Trimming size when a slow 20-day volume regime disagrees cuts the biggest losing trades, tightening the per-trade spread and lifting the in-sample Sharpe. 256 trades · mean $+4.2k/trade, median $+2.7k, win 70% · std $8.2k · skew +1.46 · in-sample Sharpe 3.44.

Base quincena

Base quincena (≤15) — per-trade net P&L distribution



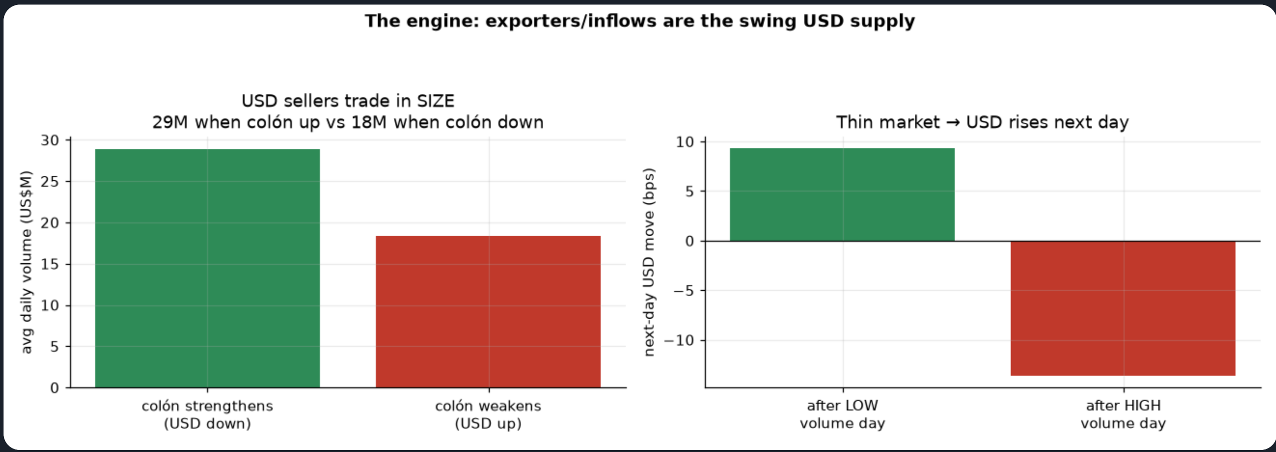
The crude ancestor: the widest spread of trade outcomes and the heaviest losers. 256 trades · mean $+3.6k/trade, median $+2.8k, win 68% · std $9.8k · skew -0.01 · in-sample Sharpe 2.05.

VARIANT · NET P&L PER TRADE, $1M/TRADE, VWAP-PRICED	TRADES	MEAN/TRADE	MEDIAN	STD	WIN %	SKEW	WORST TRADE	IS SHARPE
Real calendar (recommended)	256	$+5.1k	$+3.8k	$10.6k	71%	+0.66	$-36k	3.28
Refined (5–15)	256	$+5.1k	$+3.8k	$10.0k	72%	+0.49	$-38k	3.33
Refined + slow-vol BEST IS SHARPE	256	$+4.2k	$+2.7k	$8.2k	70%	+1.46	$-19k	3.44
Base quincena (≤15)	256	$+3.6k	$+2.8k	$9.8k	68%	-0.01	$-50k	2.05

PART B · THE UNDERLYING DYNAMICS — WHAT WE ARE EXPLOITING

B1. The engine: exporters are the swing USD supply

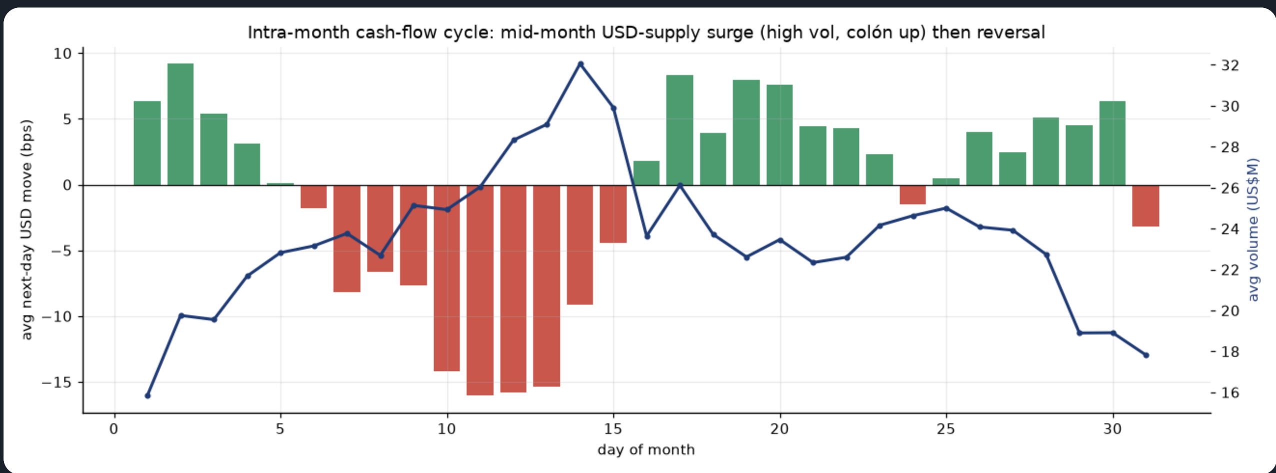
Volume when the colón strengthens vs weakens, and next-day move by volume



On days the colón strengthens (USD falls) the market trades 29M — far more than the 18M on days it weakens. USD sellers (exporters, remittances, FDI) come in size; USD buyers (importers) trickle. So HIGH volume = supply present = colón up, and LOW volume = thin market = USD drifts up next day (+9.3 bps after quiet days vs -13.7 after busy ones). That asymmetry IS the volume signal.

B2. The intra-month cash-flow cycle (why the calendar works)

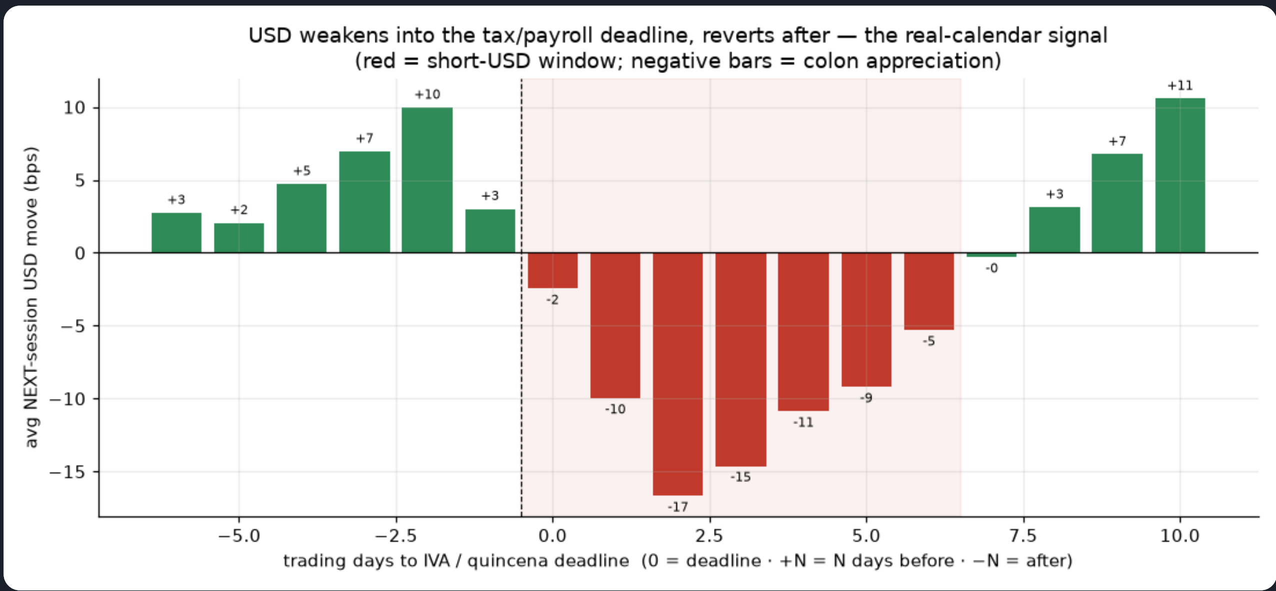
Average next-day USD move (bars) and volume (line) by day-of-month



A clear monthly rhythm: volume peaks mid-month (~day 13–15) and the colón strengthens hardest right then (red bars) — a recurring USD-supply surge (exporter settlements, tax/paydate conversions). The first days and the second half of the month lean the other way (USD up). This is the quincena effect, and it is a cash-flow cycle, not a chart pattern.

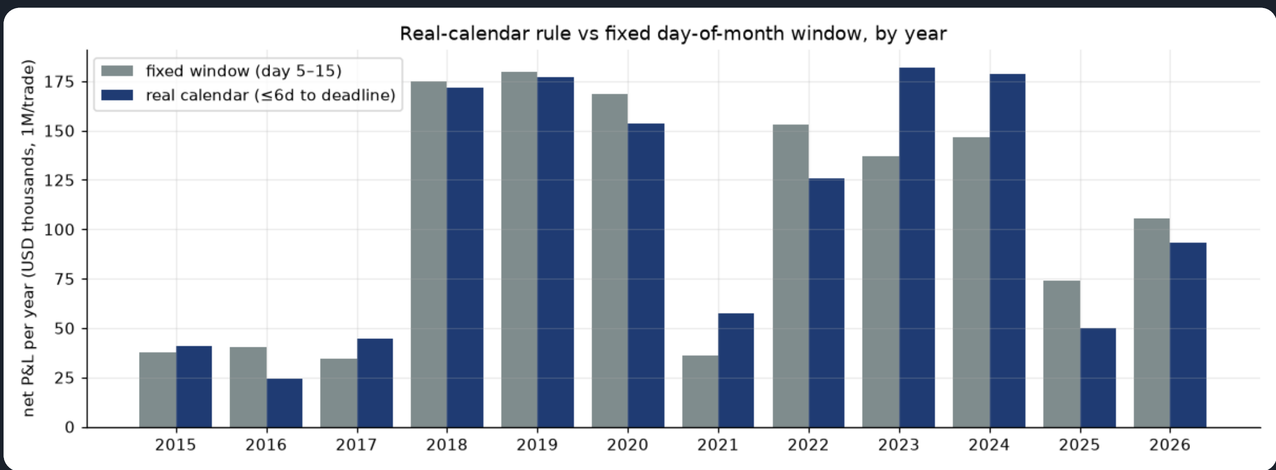
B2b. Aligning to the real deadline sharpens the signal

Average next-day USD move by business-days-to the IVA/quincena deadline



Re-indexing the same data from raw day-of-month to *trading days to the statutory IVA (15th) / mid-month payroll deadline* makes the mechanism unmistakable: the colón strengthens hardest 1–5 business days BEFORE the deadline (−12 to −16 bps next-day), then reverts to USD-up right after — a textbook conversion-then-clearing pattern. The red band is the short-USD window. Because the anchor rolls with weekends and holidays, the calendar rule (Sharpe 3.0, a broad plateau of 1.51–3.01 across lookbacks) edges the fixed-day version while explaining the flow.

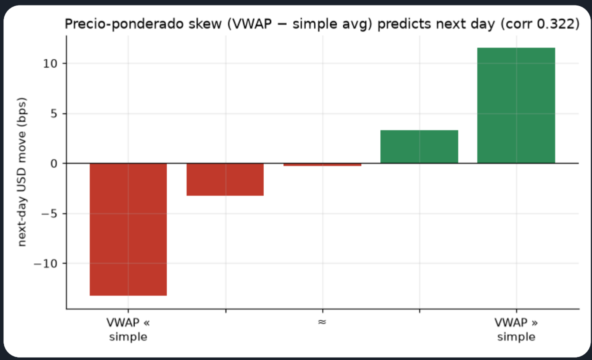
Net P&L by year — fixed day-of-month window vs real calendar ($1M/trade)



Deadline-anchoring holds up year by year (worst year $24k), not just in aggregate.

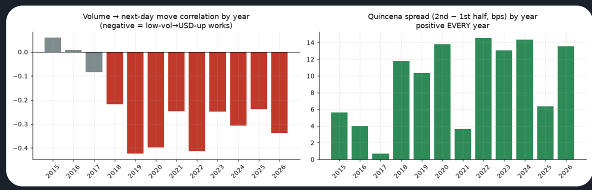
B3. Two more relationships

Precio-ponderado skew (VWAP – simple average)



When large tickets pull the weighted price above the simple average, USD tends to keep rising next day (corr 0.322). It reads large-player direction — but, as the dollar table shows, it flips daily and is too costly to trade alone.

Year-by-year stability of both effects



The volume → move link is negative every year since 2018; the quincena spread is positive in ALL 12 years. That cross-regime persistence is why we treat these as structural, not curve-fit.

PART C · HOW EACH RULE DECIDES (EXACT LOGIC)

The recommended calendar rule is the default tab; the secondary signals are one click away.

Calendar rule

Calendar / quincena rule — the recommendation

```
≤ 6 business days before the IVA/quincena deadline (through it) → SHORT USD $1M
   (firms sell USD to raise colones for the 15th tax + payroll → colón strengthens)
otherwise → LONG USD $1M (supply fades, USD drifts up)
optional: trim size to 50% when a slow 20-day volume regime disagrees
```

Why: the intra-month cash-flow cycle (Part B) — companies convert USD → CRC for the D-104 IVA filing (due the 15th, rolled to the next business day) and the mid-month quincena payroll, a recurring supply surge that strengthens the colón into the deadline and reverts after. **Two readings:** a *fixed* day-of-month window (days 1–4 LONG, 5–15 SHORT, 16–end LONG) and the *real-calendar* anchor that rolls with weekends/holidays. **Result (VWAP-priced):** $125k/yr, Sharpe 3.0, 24.2 trades/yr, max drawdown $-48k — and an out-of-sample Sharpe of 2.91 (A4). The base "short the whole first half" rule ($87k/yr, Sharpe 2.05) is the crude ancestor this refines.

Volume

Volume — "buy USD when the market is quiet"

```
seasonal_vol = today_volume ÷ (avg volume on the same weekday, to date)
if seasonal_vol < 1.0 → LONG USD $1M (quieter than normal)
if seasonal_vol ≥ 1.0 → SHORT USD $1M (busier than normal)
```

Why: quiet = exporters absent = importer demand lifts USD. Dividing by the same-weekday average makes "quiet" mean quiet *for a Tuesday*. **Result:** directionally right but flips ~77.7×/yr — even VWAP-priced it only nets Sharpe 1.1 standalone. Best used as the slow regime filter on the calendar trade.

Precio-ponderado skew

Precio-ponderado skew — "follow the big tickets"

```
if VWAP > simple_average → LONG USD $1M (large trades lifted the price)
if VWAP ≤ simple_average → SHORT USD $1M
```

Why: the weighted-vs-simple gap reveals whether large players were buying or selling USD (corr 0.322 with the next-day move). **Result:** flips ~102.2×/yr → Sharpe 0.98 net; a tie-breaker, not a standalone book.

Combined & ML

Combined vote & the ML model

```
Combined = majority vote of the three signals (always ±1) → $1M long/short
ML = logistic P(USD up) on all features; position = clip((P − 0.5) × 5, −1, +1)
```

Why the ML sizes by conviction: a raw daily flip trades ~95×/yr and dies on cost. Scaling the position by confidence keeps turnover near 59.7×/yr while still capturing the edge. On the realistic VWAP basis the combined vote nets Sharpe 2.12 and the ML 4.01 — but both lean on the same calendar + volume economics as the simple rule above.

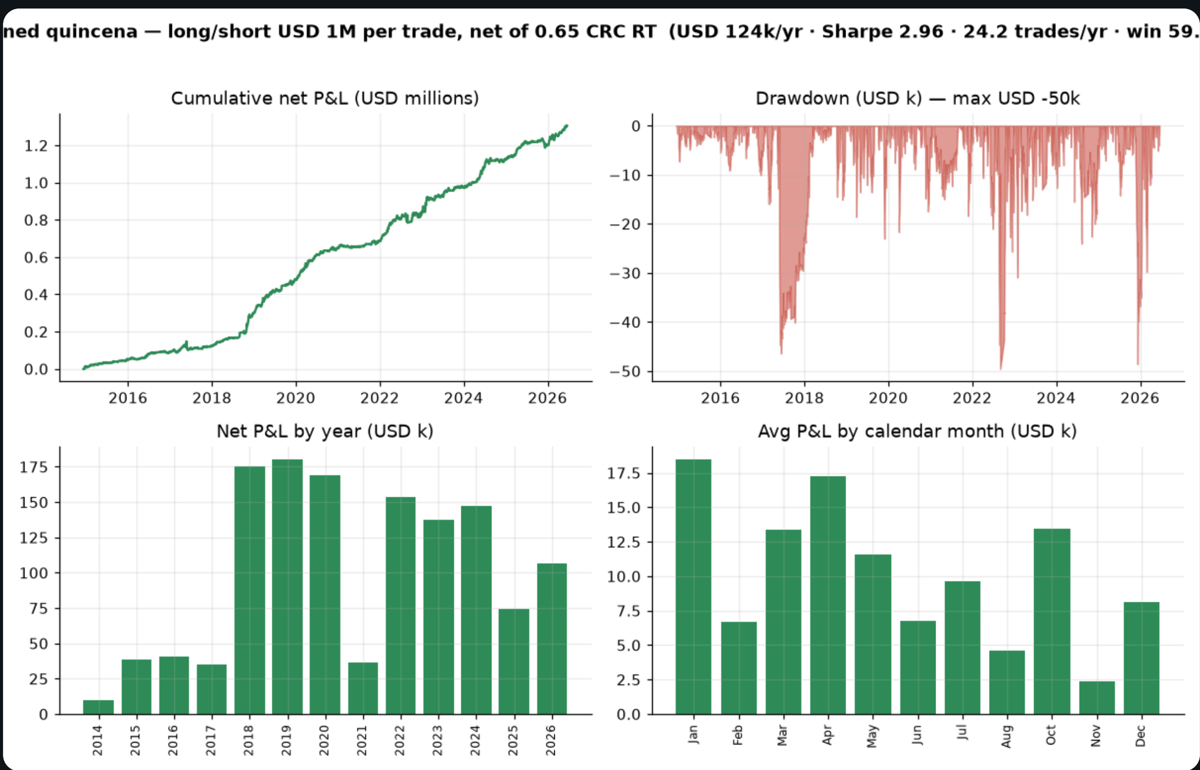
PART D · RESULTS AT $1,000,000 PER TRADE (VWAP-PRICED, NET OF COST)

The recommended calendar family sits at the top; the raw building-block signals and the ML model follow. Equity curves are tabbed — recommended first.

STRATEGY · $1M/TRADE · VWAP-PRICED, NET 0.65 CRC RT	PER YEAR	SHARPE	TRADES/YR	WIN %	MAX DD
RECOMMENDED — CALENDAR / QUINCENA FAMILY					
Real calendar (≤6 bd to deadline) BEST	$125k	3.0	24.2	59.8	$-48k
Refined quincena (short days 5–15)	$124k	2.96	24.2	59.5	$-50k
Refined + slow-vol sizing	$103k	3.37	19.9	59.2	$-37k
Base quincena (short ≤15)	$87k	2.05	24.2	56.2	$-62k
BUILDING-BLOCK SIGNALS — TRADE GENTLY / AS FILTERS					
Volume rule	$47k	1.1	77.7	50.5	$-242k
Precio-ponderado skew rule	$41k	0.98	102.2	49.5	$-397k
Combined vote (3 signals)	$89k	2.12	85.2	53.6	$-278k
ML combined model (conviction-sized, walk-forward)	$134k	4.01	59.7	52.3	$-36k

Recommended calendar

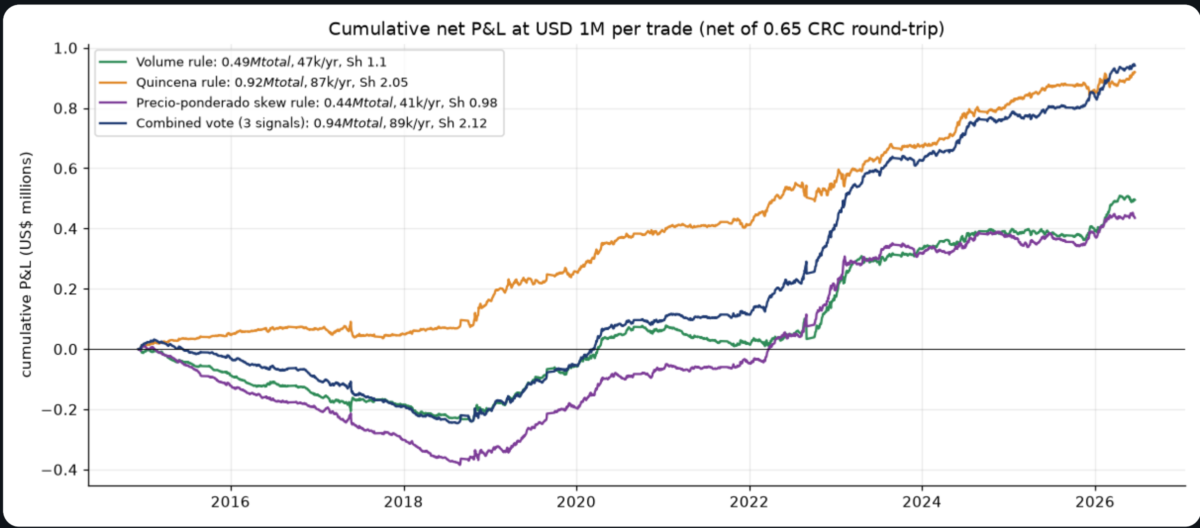
Recommended calendar rule — equity, drawdown, yearly & monthly P&L



Smooth compounding at ~$125k/yr with shallow drawdowns; every year and (on average) every month positive — January and April strongest. VWAP-priced, net of cost.

Building-block rules

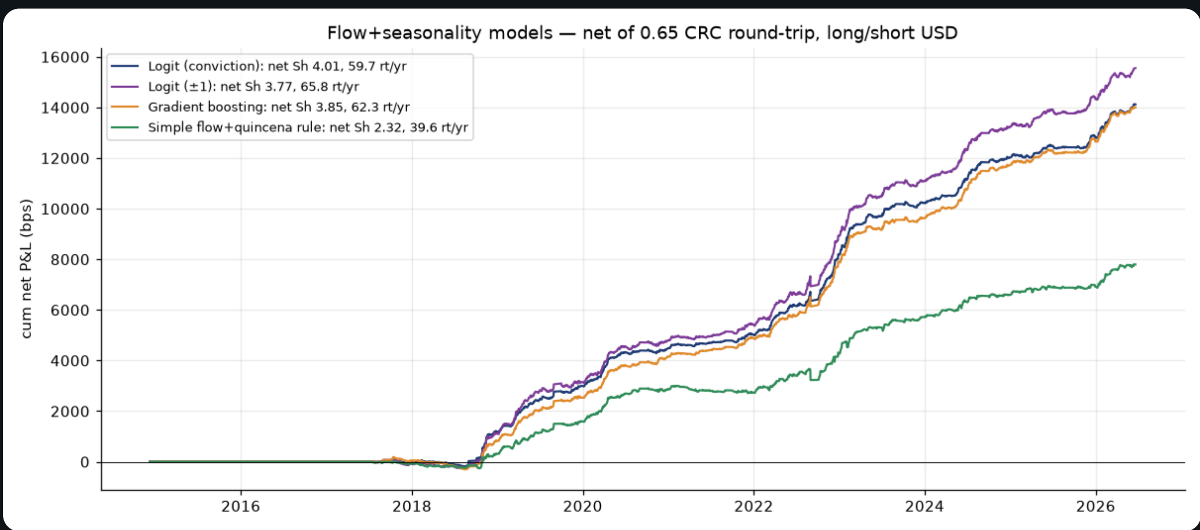
Cumulative net P&L in US$ millions — building-block rules, $1M per trade



The base quincena (green) compounds smoothly; the combined vote is decent but dragged by the turnover of the volume and skew legs. The volume rule is directionally right but churns — the signal is real, the daily flipping is the cost problem.

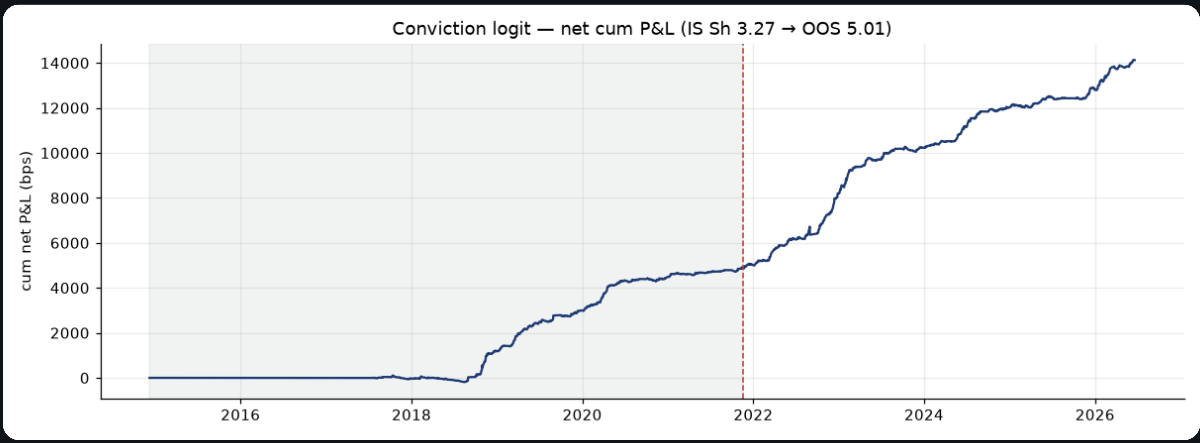
ML model

Conviction-sized ML model — cumulative net P&L (walk-forward, VWAP-priced)



Every prediction is out-of-sample (walk-forward). Conviction sizing keeps turnover near 59.7×/yr; net Sharpe 4.01. It mostly repackages the same calendar + volume economics as the simple rule, at higher complexity.

ML: in-sample vs out-of-sample (net)

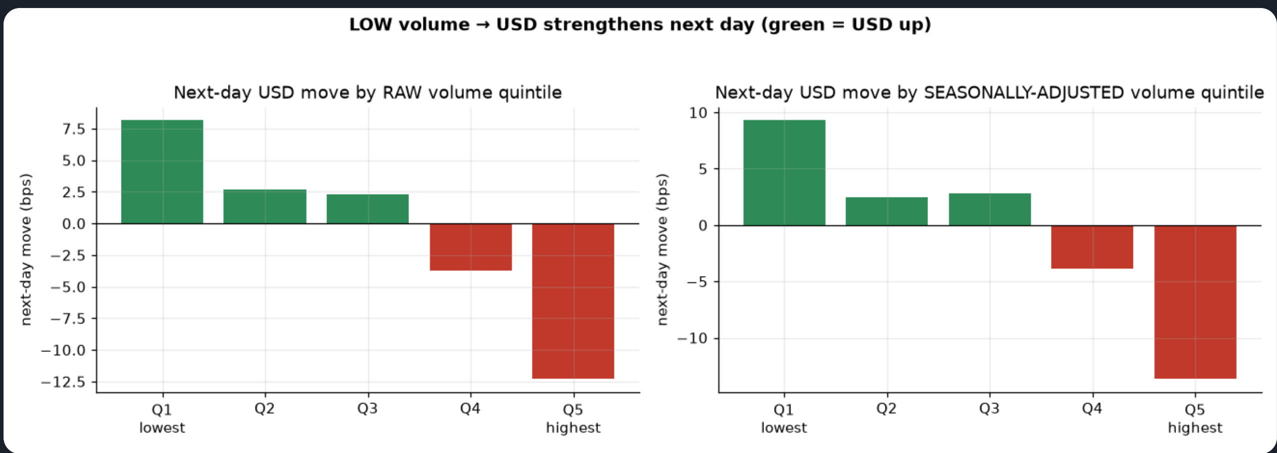


OOS Sharpe 5.01 vs IS 3.27; a 5-day feature lag collapses accuracy (no look-ahead), permutation p=0.0.

PART E · RAW RELATIONSHIPS BEHIND THE SIGNALS

E1. Low volume → USD up (seasonally adjusted)

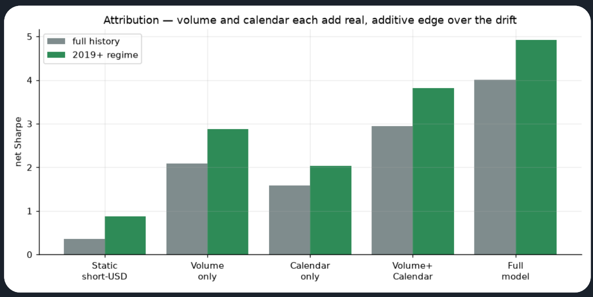
Next-day USD move by volume quintile



Monotonic: quietest days → USD up, busiest → USD down. Seasonal adjustment sharpens the extremes. This is the raw relationship the volume filter and the ML model lean on.

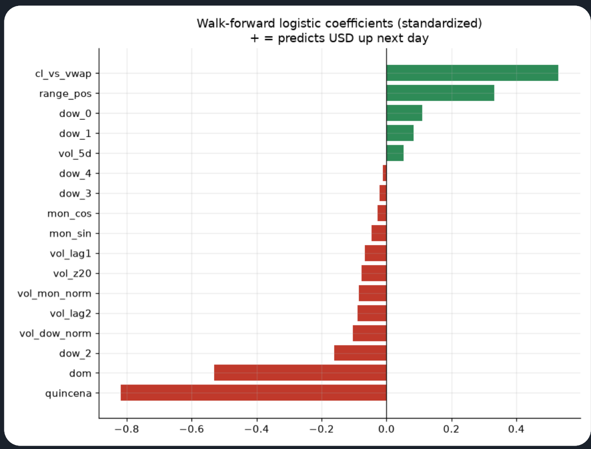
E2. Attribution & drivers

Net Sharpe by feature family



Volume-only and calendar-only each beat the drift and are additive — the calendar carries the most, which is why the standalone calendar rule is the recommendation.

Walk-forward model coefficients



Quincena/day-of-month and volume dominate; no technical indicators used.

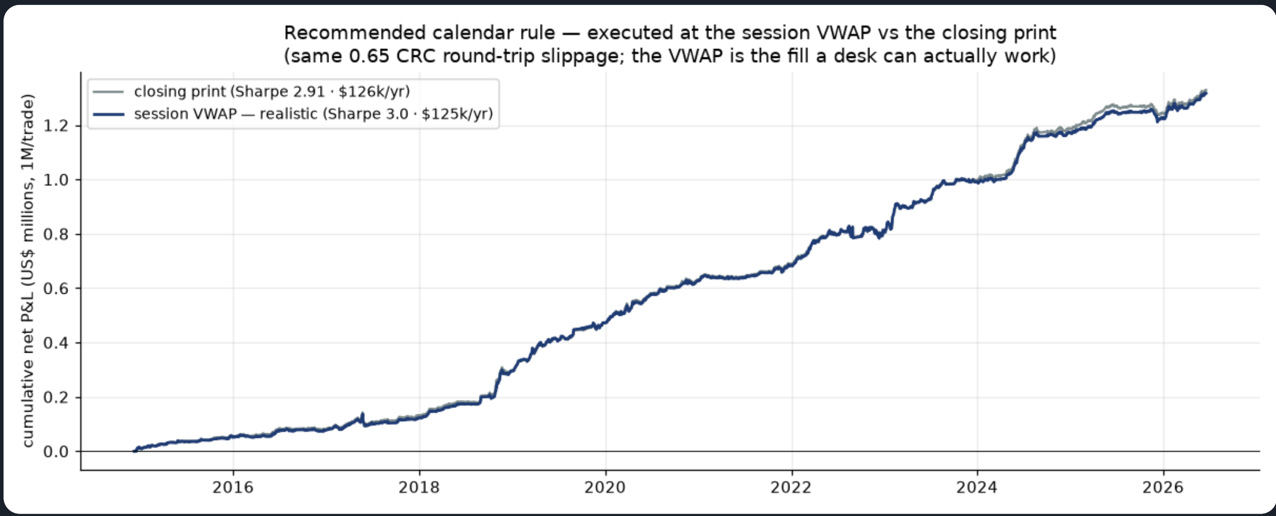
PART F · EXECUTION REALISM — VWAP VS THE CLOSING PRINT

F1. The recommended rule does not depend on catching the close

Every result in this report is already priced **VWAP-to-VWAP** — the fill a desk can actually work by spreading an order across the session, rather than the optimistic assumption of trading the last print. Here is the proof for the strategy we care about: the same calendar rule, same 0.65 CRC round-trip slippage, marked at the closing print vs the session VWAP. The VWAP fill **does not haircut the edge — it slightly improves the Sharpe** (2.91 → 3.0), because the VWAP is a steadier reference than a single closing tick.

RECOMMENDED CALENDAR RULE · $1M/TRADE · NET 0.65 CRC RT	PER YEAR	SHARPE	MAX DD
Closing-print fill (optimistic mark)	$126k	2.91	$-45k
Session VWAP fill USED EVERYWHERE IN THIS REPORT	$125k	3.0	$-48k

Recommended calendar rule — session VWAP vs closing-print execution



Both curves use the identical signal and the identical 0.65 CRC round-trip slippage; only the fill reference differs. They track each other closely and the VWAP (realistic) curve carries the higher Sharpe — the calendar edge is an economic flow, not an artifact of marking to the close.

F2. Secondary strategies at VWAP (kept for completeness)

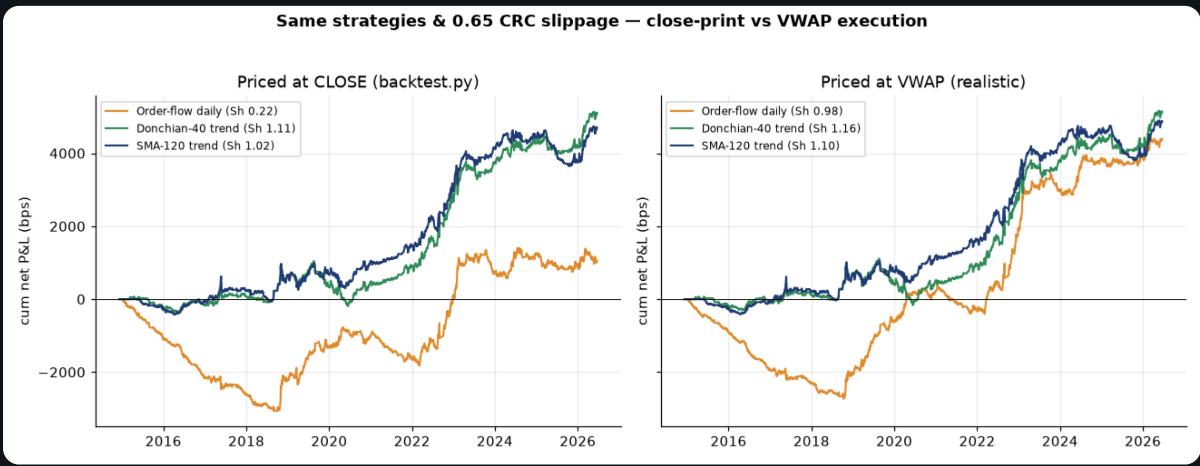
Why they're here

These trend / order-flow books are **not** the recommendation — they are shown only to demonstrate that the VWAP-execution finding generalises beyond the calendar rule. On the realistic VWAP basis the noisy daily order-flow book improves most (its turnover is where a single closing tick hurts), while the slow trend books barely move. If you only care about the recommended strategy, F1 above is the whole story.

SECONDARY STRATEGY (NET 0.65 CRC RT, VWAP-TO-VWAP)	CLOSE SHARPE	VWAP SHARPE	VWAP BPS/YR	TRADES/YR
Order-flow daily	0.22	0.98	415	95.2
Donchian-40 trend	1.11	1.16	486	2.5
SMA-120 trend	1.02	1.1	461	7.0

Close vs VWAP chart

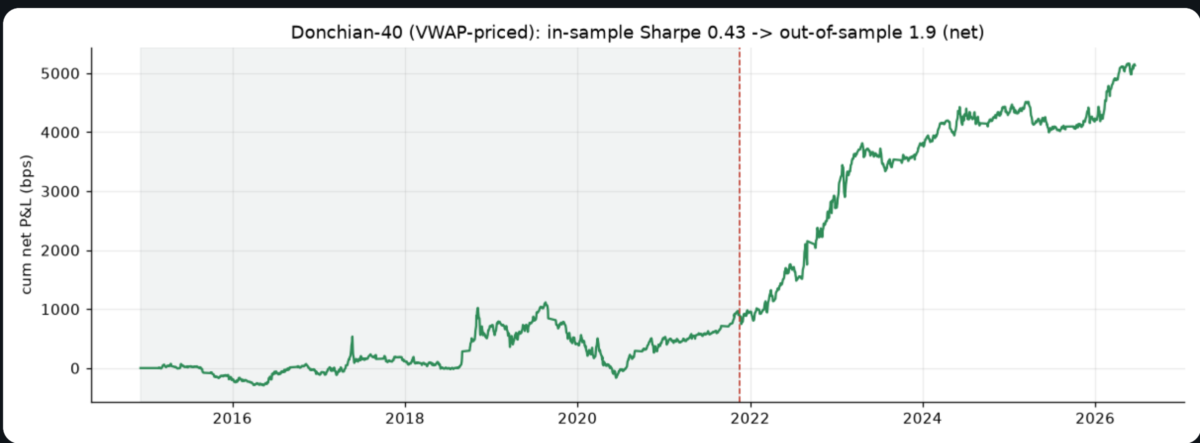
Secondary books & slippage — close-print vs VWAP execution



Left priced at the close, right at the VWAP. The trend books (Donchian-40 1.11→1.16, SMA-120 1.02→1.1) barely move; the daily order-flow book improves most (0.22→0.98).

Out-of-sample

Donchian-40 (VWAP-priced): in-sample vs out-of-sample net Sharpe



Held out 60/40, the VWAP-priced Donchian-40 keeps working — OOS net Sharpe 1.9 (≥ IS 0.43).

PART G · VERDICT

The recommendation is the calendar rule. Short USD into Costa Rica's mid-month IVA / payroll deadline (within 6 business days of it, or day-of-month 5–15), long the rest of the month — **$125k/yr per $1M at Sharpe 3.0**, VWAP-priced, 24.2 trades/yr, worst year $34k, positive in all 12. A slow 20-day volume filter lifts the risk-adjusted return to Sharpe 3.37 (drawdown just $-37k). Start here.

A dynamic exit enhances it (A5). Because the calendar edge front-loads and reverts, a 80-bps **trailing stop** that banks each trade once the move stalls raises P&L to $128k/yr (Sharpe 3.22) — **up in both the in-sample and out-of-sample windows**. Adding a hard 60-bps floor takes the Sharpe to 3.91 and nearly halves the worst drawdown ($-48k → $-26k). Both were tuned on in-sample data only, and the whole 30–100-bps band works — it is an overlay on the calendar rule, not a replacement.

It holds out of sample. On a chronological 60/40 split (test window after 2021-11-18, nothing re-fit) the rule earns $152k/yr at Sharpe 2.91 out-of-sample — if anything stronger than the 3.28 in-sample, because the post-2021 float is where the cash-flow cycle is most tradeable.

Your volume thesis is correct but must be traded gently. Low volume → USD up is real and monotonic, but a daily flip pays much of it to slippage even VWAP-priced. Use it conviction-sized or as the slow filter on the calendar trade — that is what lifts the combined/ML book to Sharpe 2.12–4.01.

The mechanism is economic, not technical: exporter USD supply (volume) and the payment/tax cycle (quincena). That is why it persists across regimes — and why it would weaken if the BCCR re-pegged or intervened heavily, the dominant risk.

Priced realistically, the edge survives. Everything here is marked at the session VWAP, not the closing print. For the recommended rule that is a slight *improvement* (2.91 → 3.0 Sharpe), so the result is not an artifact of marking to the close.

Dollar & pricing convention. $1,000,000 USD notional per full position; 1 bp of next-session move = $100; slippage 0.325 CRC/side (~$680) charged on every position change. **All P&L is priced VWAP-to-VWAP** — a position decided from information through session *t* earns the next session's VWAP move, and the slippage is booked against the VWAP (the realistic desk fill), net, gross of financing/borrow and market impact, non-compounded (reset to $1M each day). The ML row is conviction-sized so its notional varies up to $1M; all model numbers are walk-forward out-of-sample. In-sample / out-of-sample = chronological 60/40 split at 2021-11-18. Reproducible: `parse_monex → analyze → eda → volume_model → dynamics → quincena → exits → backtest_vwap → build_report`.